

When Is an Inverse a Function?

Algebra II | Unit 1 | Day 12 | TEK A2.2.C

Objective: I can decide whether the inverse of a function is itself a function, and restrict the domain of the original when it is not.

Vocabulary

- Vertical line test** — if any vertical line hits the graph twice, it is _____.
- One-to-one** — every output comes from exactly _____ input. Test with a _____ line.
- Key fact** — every inverse of a function is a _____, but it is a _____ only when the original is one-to-one.
- Domain restriction** — deliberately shrinking the domain so the function _____.

Part 1 — The Problem with $f(x) = x^2$

Fill in the table, then look for repeats in the bottom row.

x	-3	-2	-1	0	1	2	3
$f(x) = x^2$							

- Which outputs appear more than once? _____ So f is _____ one-to-one.
- Now swap to build the inverse. Both $(1, 1)$ and $(1, -1)$ appear. Is that a function? _____
- Why not? _____
- Written out, the inverse relation is $y =$ _____, and the $\pm$ is exactly the problem.

Part 2 — Fix It with a Restriction

Keep only half of the parabola, so no output repeats.

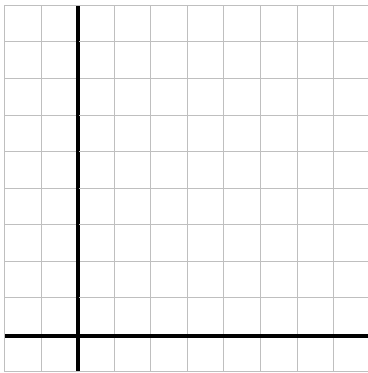
Restriction on $f(x) = x^2$	Which half of the parabola	Then $f^{-1}(x) =$
$x \geq 0$	_____	_____
$x \leq 0$	_____	_____

- Either restriction works. The usual choice is _____, which gives the familiar $f^{-1}(x) = \sqrt{x}$.
- Restricting the domain of f also restricts the _____ of f^{-1} .

Part 3 — Graph the Restricted Pair

Each square is 1 unit. Graph only the half of x^2 you kept, then its inverse, plus $y = x$.

$$f(x) = x^2 \text{ for } x \geq 0, \text{ and } f^{-1}(x) = \sqrt{x}$$



Record it.

Domain of restricted f: _____

Range of restricted f: _____

Domain of f^{-1} : _____

Do they mirror across $y = x$? _____

Part 4 — Which Pairs Need No Restriction?

Some parent functions are already one-to-one, so their inverses are functions with no help.

Function	One-to-one?	Restriction needed?	Its inverse
$f(x) = x^2$	_____	_____	_____
$f(x) = x^3$	_____	_____	_____
$f(x) = 2^x$	_____	_____	_____
$f(x) = 2x + 1$	_____	_____	_____
$f(x) = x $	_____	_____	_____

Pattern: functions that only _____ (or only decrease) are one-to-one. Ones that turn around, like x^2 and $|x|$, are not.

Part 5 — Quick Check

1. Is the inverse of every function also a function? _____
2. Which test tells you the inverse will be a function? _____
3. Why does x^3 need no restriction but x^2 does? _____

Exit Ticket

Consider $f(x) = x^2 - 4$ with no restriction.

Is its inverse a function? _____ Give a restriction that fixes it: _____

Explain in one sentence why the restriction is needed: _____